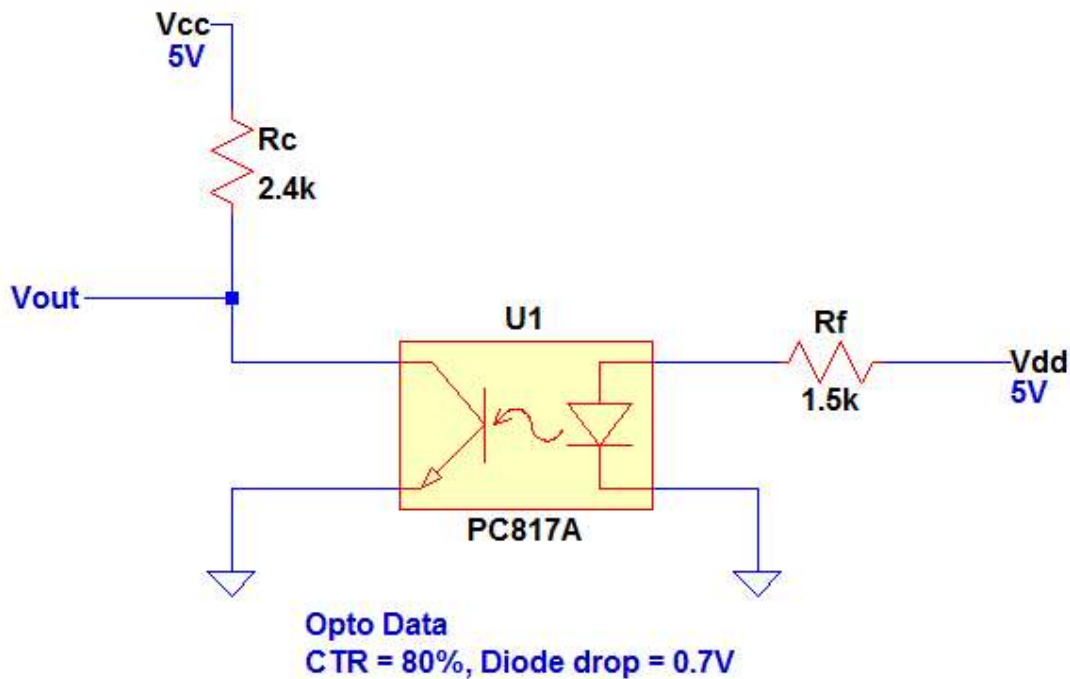




Electronics Optocoupler

Optocoupler Circuit Design and Detailed Analysis

by [electronicsbeliever](#) • 2 • Electronics, Optocoupler

Actually, Optocoupler circuit design is not that difficult as some thought. It's just like you are designing a BJT circuit. If a BJT has its beta or current gain, optocoupler has its CTR or current transfer ratio. Once you know what a CTR is and learn how to use it, then Optocoupler circuit design is that easy.

Current transfer ratio or just CTR is the ratio of the collector to the forward current which is expressed in percent.

$$\text{CTR} = (I_c / I_f) \times 100\%$$

Collector current is the current that will flow to the collector of the transistor side of the optocoupler. On the other hand, the forward current is the current that flow to the diode side of the optocoupler.

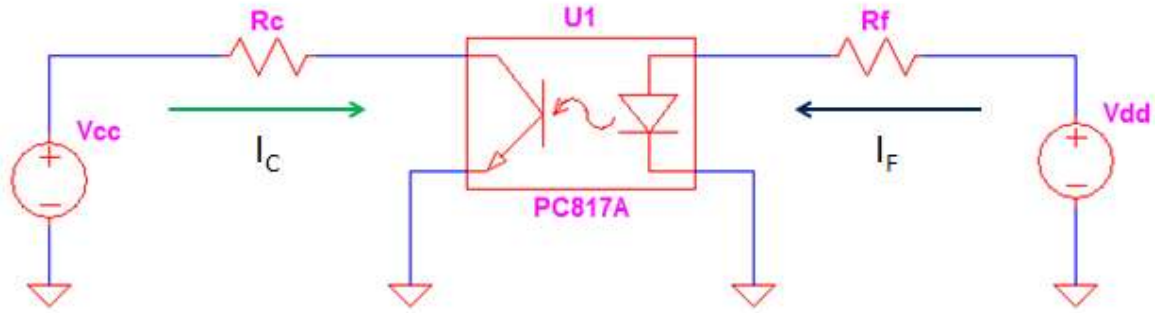


Figure 1

Basically the diode side is linked to the transistor side by the device current transfer ratio. Apart from this information, Optocoupler circuit design is just like designing ordinary circuit wherein you are using KVL, KCL, ohms law and so on.

Optocoupler Circuit Design Steps

1. Select a Circuit Structure

Do not complicate your circuit. The lesser the parts count the better in two different ways. First reason is of course a lesser cost. The other reason is that with a lesser parts, the lower the total failure rate and the higher the circuit reliability.

Supposing you are only interested to transmit signal from primary side controller to the secondary controller, your circuit must be simple as below.

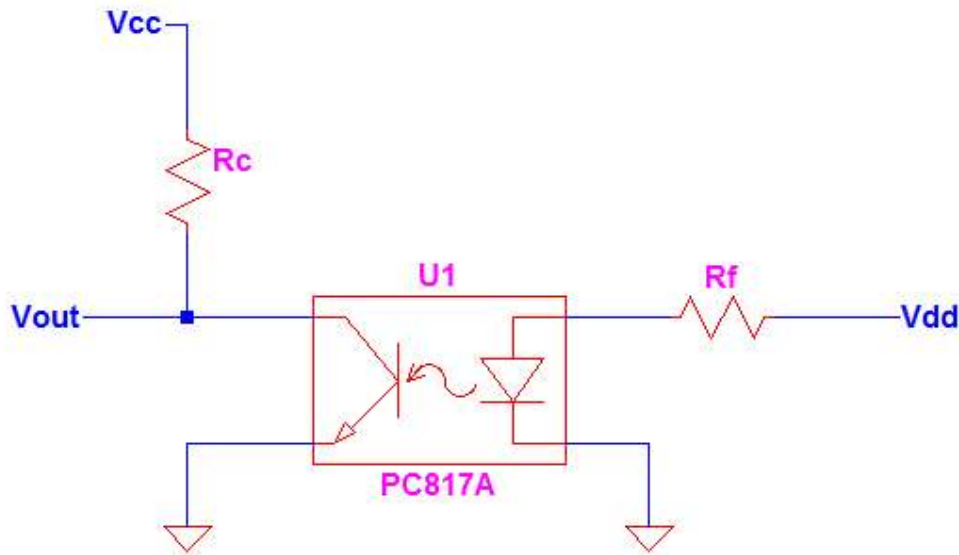


Figure 2

Above circuit configuration is an inverter. If you want a non-inverting signal, you can use below structure.

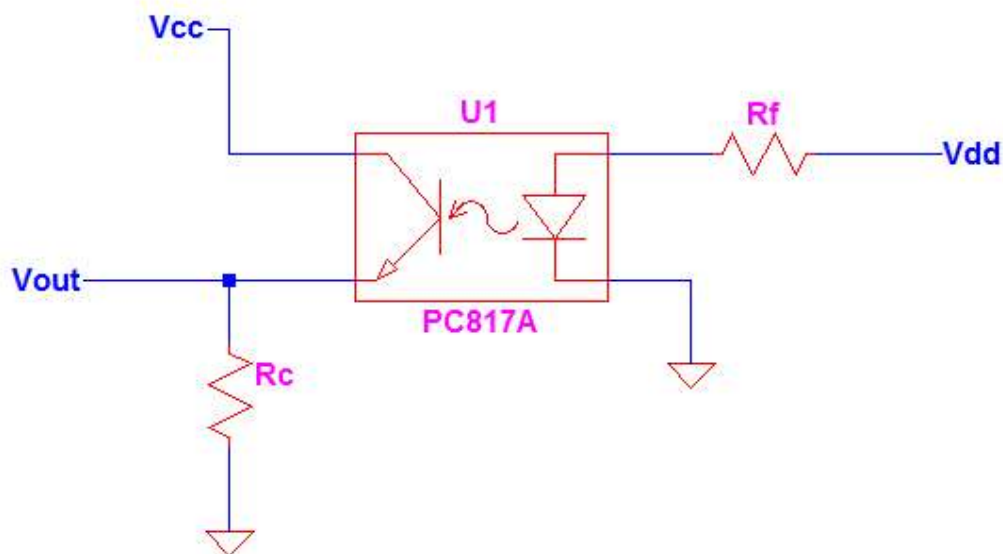


Figure 3

The first circuit is normally an inverting one if you are going to saturate the transistor. However if you bias the circuit to operate in the linear region, you can get a voltage higher than zero at V_{out} node. The second circuit is a non-inverting configuration which is comparable to a common collector configuration of a BJT. But BJT common collector is more complicated than this circuit with the presence of the base current.

2. Select an Optocoupler Part

The next step in Optocoupler circuit design is to select an optocoupler part. In doing so, you must consider your application. If your application is a switch, you must select a device with a higher minimum CTR. If your application is linear, you can consider using the one with a tight CTR range. A tight CTR will correspond to a smaller variation.

If the design is to be exposed in high surrounding temperatures, you better select an optocoupler with a CTR that will not much affected with the ambient temperature. Optocoupler CTR will decrease with increasing temperatures. For example in below graph, the relative CTR will decrease much at 100°C ambient temperature.

Fig. 7 Relative Current Transfer Ratio vs. Ambient Temperature

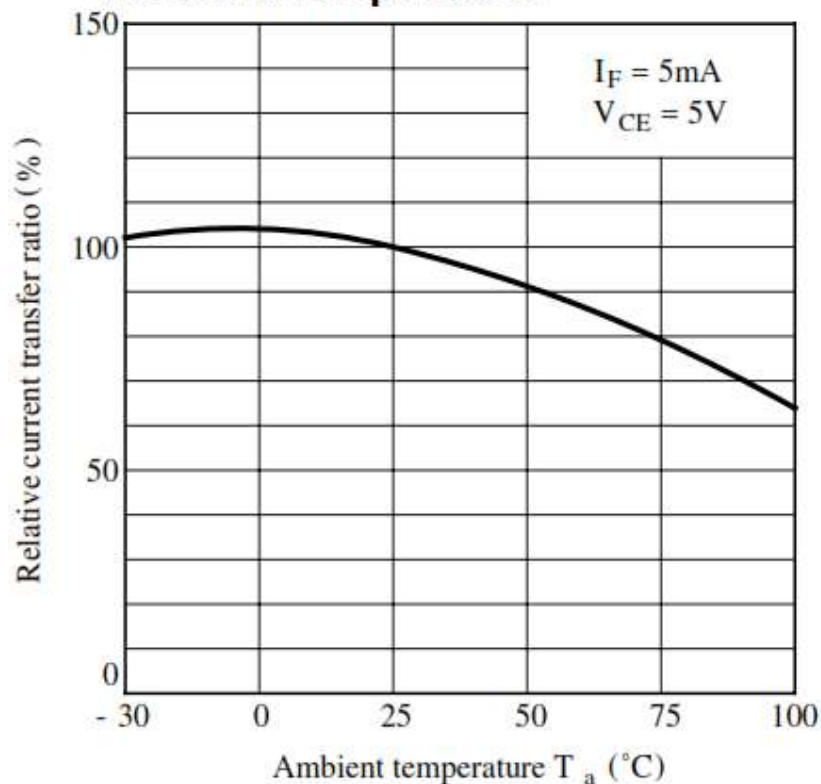


Figure 4

For long term product operation, you better consider also the life expectancy curve. As you see in the graph below, CTR will decrease as the device got aged.

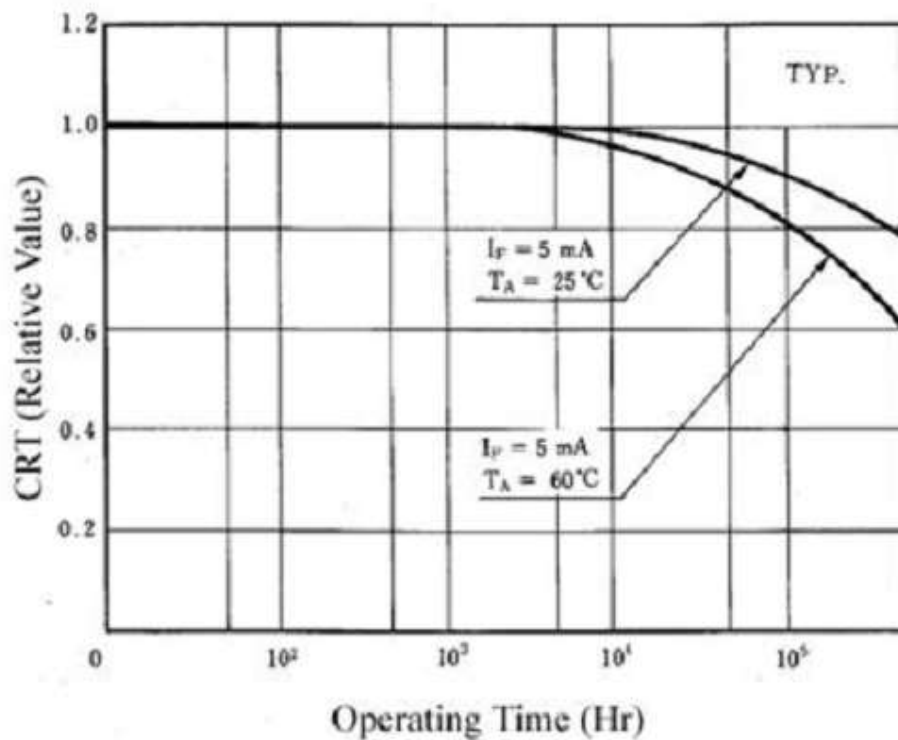


Figure 5

3. Set Circuit Operation

This time you have to set the operating point of the circuit. To make this more informative, we are going to refer to the circuit below.

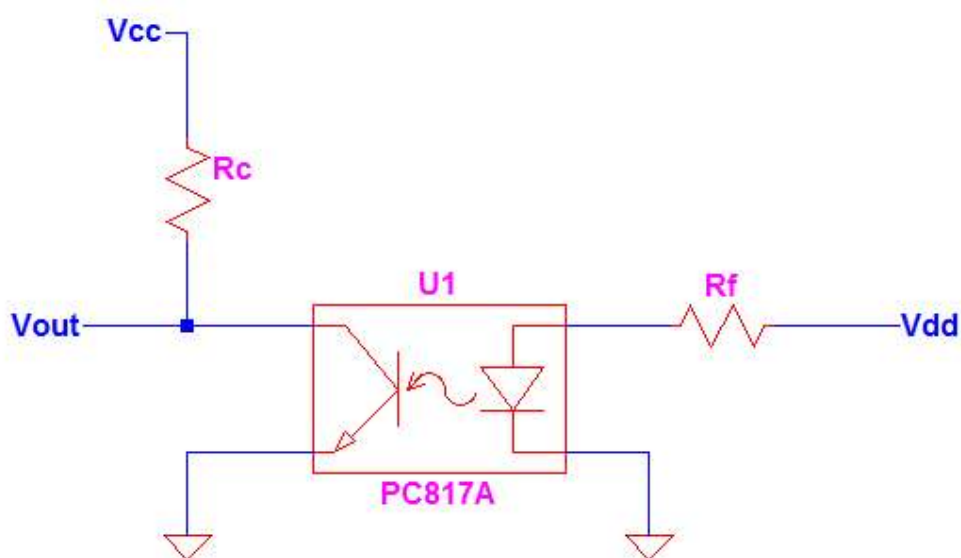


Figure 6

Define Output Level

The above circuit is can be configured to operate at linear or saturation regions. At saturation the Vout node is zero ideally while above zero but lower than Vcc in linear. When there is no bias in the diode side the level of Vout is simply the same to Vcc. So if you design the circuit as a switch, you must ideally assume a zero VCE or Vout when the Optocoupler is conducting. If the application is linear, you must define a particular level in the Vout node to be used in the design.

Define Rf Value

You can freely choose this value. However, in some applications you need to be careful. Most of the times, Vdd is derived from a digital circuit or device such as MCU or DSP. If so, set the value of Rf in such a way the current rating of the digital circuit or device is not exceeded. For MCU and DSP the [sink and source](#) currents are usually ranging from 4mA to 9mA (some other may reach higher than 9mA, you can always check in the datasheet). Supposing the current rating is only 4mA maximum, set actual forward current to at most 80% of it. So Rf would be

$$I_f = (V_{dd} - V_f) / R_f$$

$$R_f > (V_{dd} - V_f) / (80\% \times I_{rating})$$

(Vf is the forward voltage of the optocoupler)

Determine Rc

After defining Rf and the forward current, you can determine the value of Rc.

For Saturation Setting

$$(I_c / I_f) < CTR_{device}$$

$$I_c = (V_{cc} - V_{CEsat}) / R_c$$

$$R_c > [(V_{cc} - V_{CEsat}) / (CTR_{device} \times I_f)]$$

For Linear Setting

$$(I_c / I_f) = CTR_{device}$$

$$I_c = (V_{cc} - V_{CE}) / R_c$$

$$R_c = [(V_{cc} - V_{CE}) / (CTR_{device} \times I_f)]$$

Design Example

Applying the Optocoupler circuit design steps above will make the discussion more informative. Now, let us provide values for the circuit below. The output should provide logic low and logic high level. A logic low level is any voltage below 0.8V while a logic high level is equal to V_{cc} . The supply V_{cc} is 5V provided by an MCU with a source and sink current capability of 4mA. The optocoupler CTR is 80% and the diode drop is 0.7V. Consider V_{cc} of 5V.

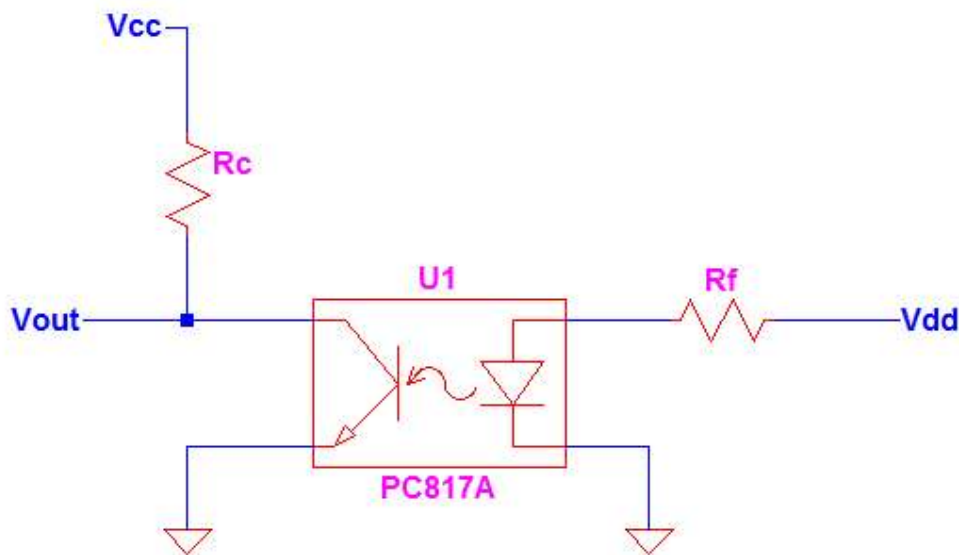


Figure 7

Select R_f Value

$$R_f = [(V_{dd} - V_f) / (80\% \times I_{rating})]$$

$$R_f > [(5V - 0.7V) / (80\% \times 4mA)] = \mathbf{1.34 \text{ kohm}}$$

We can set R_f to 1.5k standard value.

Solve for I_f

$$I_f = [(5V - 0.7V) / 1.5 \text{ kohm}] = \mathbf{2.87 \text{ mA}}$$

Determine R_c

$$R_c > [(V_{cc} - V_{CEsat}) / (CTR \times I_f)] = [(5V - 0V) / (80\% \times 2.87\text{mA})] = \mathbf{2.18 \text{ kohm}}$$

We assume V_{CEsat} equals zero for worst case.

Now the final circuit with values is

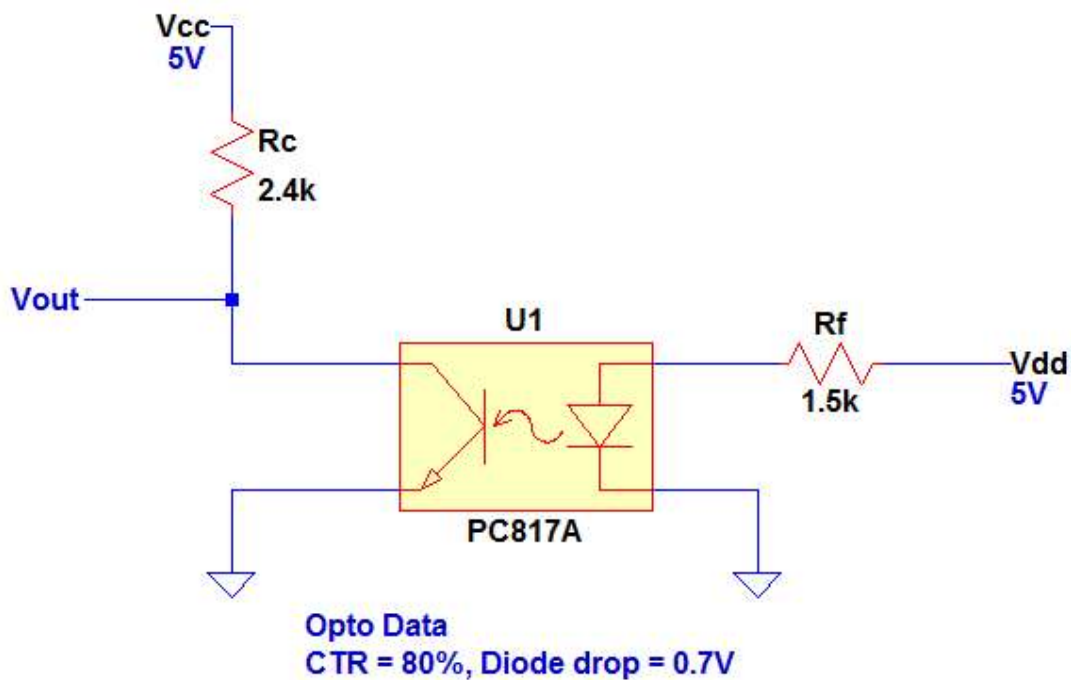


Figure 8

Design Check

We will verify if the forward current I_f is not exceeding the maximum source and sink current of the MCU.

$$I_f = [(5V - 0.7V) / 1.5 \text{ kohm}] = \mathbf{2.87 \text{ mA}}$$

The computed current forward current is safe.

Check if the Optocoupler Can Output a Low Signal

To attain a low signal, the transistor side must saturate. To know if the transistor can really saturate we use below relation.

$$(I_c / I_f) < CTR_{\text{device}}$$

Compute of the collector current during saturation

$$I_c = 5V / 2.4 \text{ kohm} = \mathbf{2.08 \text{ mA}}$$

$$I_c / I_f = 2.08 \text{ mA} / 2.87 \text{ mA} = \mathbf{72.47 \%}$$

The device CTR is 80%, so the transistor can saturate. In order to ensure a hard saturation, you can add more margins to the collector resistor; say by adding 50% to the computed value.

How about a high logic, will the circuit deliver? The logic high is not a problem because once Vdd is remove the transistor will cutoff, Vout node will see Vcc level

Another scenario where the operation is linear

Provide circuit values so that the Vout node will have 3V level. Use same supply levels and other given with the previous example.

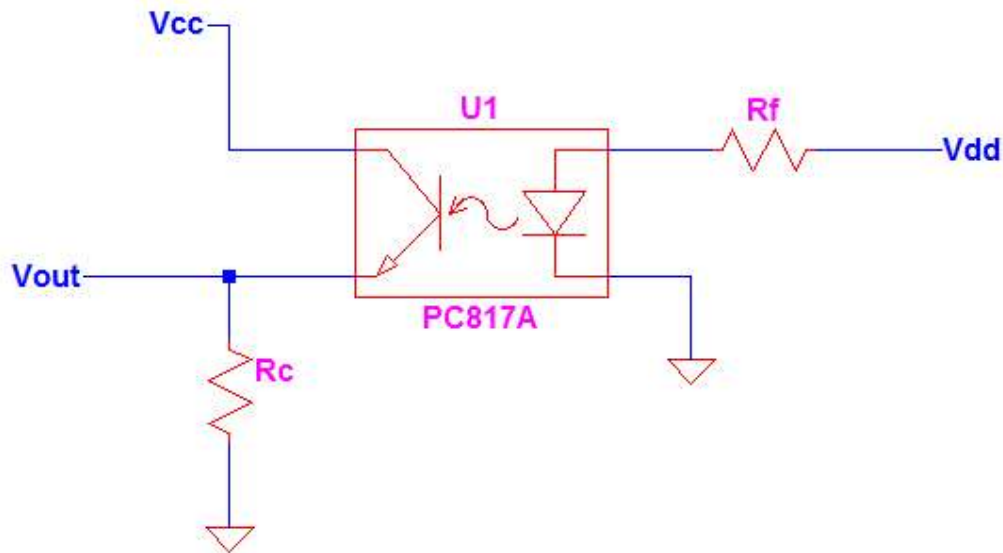


Figure 9

Select Rf Value

$$R_f > [(V_{dd} - V_f) / (80\% \times I_{rating})] = [(5V - 0.7) / (80\% \times 4 \text{ mA})] = \mathbf{1.34 \text{ kohm}}$$

We can set Rf to 1.5k standard value.

Solve for If

$$I_f = [(5V - 0.7V) / 1.5 \text{ kohm}] = \mathbf{2.87 \text{ mA}}$$

Determine Rc

$$R_c = [(V_{cc} - V_{CE}) / (CRT \times I_f)]$$

The specified level of Vout is 3V which makes the VCE equal to 2V, so

$$R_c = [(5V - 2V) / (80\% \times 2.87 \text{ mA})] = \mathbf{1.31 \text{ kohm}}$$

Use a standard value very close to the computed value. In this case we will use 1.3kohms.

Checking

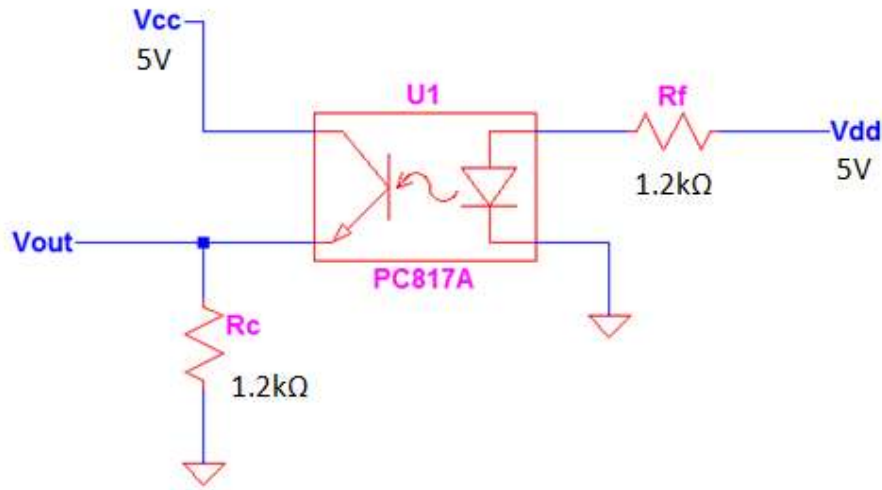


Figure 10

$$I_f = [(5V - 0.7V) / 1.5 \text{ kohm}] = \mathbf{2.87 \text{ mA}}$$

$$I_c = \text{CTR} \times I_f = 80\% \times 2.87 \text{ mA} = \mathbf{2.296 \text{ mA}}$$

$$V_{out} = I_c \times R_c = 2.296 \text{ mA} \times 1.3 \text{ kohm} = \mathbf{2.984 \text{ V}}$$

The resulting V_{out} is not exactly equal to 3V because we are using 1.3k for the value of R_c instead of the computed 1.31k.

You can repeat the steps above in any Optocoupler circuit design. Once you use the techniques often, the task will become very easy. Another thing that makes it easier is that there is no base current as with BJT.

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2 thoughts on “Optocoupler Circuit Design and Detailed Analysis”

Anand Paralkar July 8, 2016 at 3:43 AM

Thank you for your detailed analysis of the optocoupler. It is by far the most informative site I have seen on this neglected part of the power industry.

Could you please tell me if there is reliable way to get the PC817 to operate in the linear region? I am asking this because I see that the CTR varies a lot for the PC817A/B/C/D optocouplers.

Would it be better to use something like the IL300?

My application is to use an optocoupler along with a TL431 in the control loop of forward converter.

REPLY

jlsoronio August 31, 2016 at 9:49 AM

I agree that opto CTR is wide. For tighter requirement, select an optocoupler which CTR range is narrow.

For opto in the loop, it is very recommended to use a very tight CTR. If you want PC817 part, use the “A” version as it has very narrow CTR range.

Optos are usually linear once operated in the CTR range. However, you need to consider temperature variation. Select

an opto with CTR is not well affected with the temperature, see the graph on the datasheet.

Read this for more info:

<https://electronicsbeliever.com/how-to-determine-optocoupler-operation-saturation-or-linear/>

REPLY

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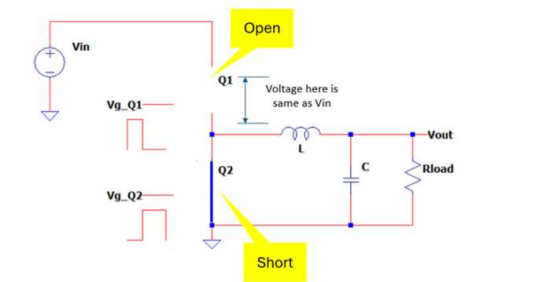


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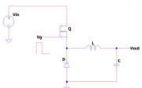
Quick Facts About Synchronous Buck Converter

Actually, Optocoupler circuit design is not that difficult as some thought. It's just like you are designing a BJT circuit. ... [Read more](#)

SUPPLY VOLTAGE		
Parameters	Value	Unit
VGG	5	volt
VDD	5	volt
This is the typical value of VGG. See the figure.		
This is the typical value of VDD. See the figure.		
AMBIENT TEMPERATURE		
Parameters	Value	Unit
Tamb_max	60	°C
Tamb_min	-30	°C
This is the maximum operating ambient temperature. This is a design given.		
This is the minimum operating ambient temperature. This is a design given.		
RESISTOR DATA		
Parameters	Value	Unit
R01	1500	ohms
R02	1000000	ohms
ps_R01_rating	0.1	watt
ps_R02_rating	0.1	watt
Input here the resistance of R01. See figure.		
Input here the resistance of R02. See figure.		
Input here the power dissipation rating of R01. Get this from the datasheet.		
Input here the power dissipation rating of R02. Get this from the datasheet.		
RELAY DATA		
Parameters	Value	Unit
COIL_typical	100	ohms
LOCL	100	uH
ps_coil_rating	1	watt
Tmin_relay	-40	°C
Tmax_relay	125	°C
Input here the coil resistance of the relay. Get this from the datasheet.		
Input here the coil inductance of the relay. Get this from the datasheet.		
Input here the power dissipation rating of the relay coil. Get this from the datasheet.		
Input here the minimum operating temperature of the relay. Get this from the datasheet.		
Input here the maximum operating temperature of the relay. Get this from the datasheet.		

Design Template for NMOS Lowside Relay Driver

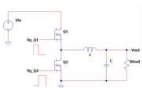
Actually, Optocoupler circuit design is not that difficult as some thought. It's just like you are designing a BJT circuit. ... [Read more](#)



Buck Converter Design Template
Notes:
- This is a design template. The user must enter values for the components.
- The user must enter values for the components.
- The user must enter values for the components.
- The user must enter values for the components.
- The user must enter values for the components.

Buck Converter Design Using Excel Calculator

Actually, Optocoupler circuit design is not that difficult as some thought. It's just like you are designing a BJT circuit. ... [Read more](#)



Synchronous Buck Converter Design Template
Notes:
- This is a design template. The user must enter values for the components.
- The user must enter values for the components.
- The user must enter values for the components.
- The user must enter values for the components.
- The user must enter values for the components.

Synchronous Buck Converter Design Calculator Excel-External Switches

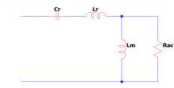
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Parameter	Symbol	Value	Unit
Input Voltage			
Minimum input voltage	V_{in_min}	300	V
Nominal input voltage	V_{in_nom}	360	V
Maximum input voltage	V_{in_max}	400	V
Output Voltage			
Minimum output voltage	V_{out_min}	15	V
Nominal output voltage	V_{out_nom}	24	V
Maximum output voltage	V_{out_max}	30	V
Load Current			
Minimum load current	I_{load_min}	0.1	A
Nominal load current	I_{load_nom}	0.2	A
Maximum load current	I_{load_max}	0.3	A
Frequency Targets			
Resonant Frequency	f_r	100	kHz
Switching Frequency	f_s	200	kHz
Output Power			
Minimum output power	P_{out_min}	300	W
Nominal output power	P_{out_nom}	360	W
Maximum output power	P_{out_max}	400	W
Transformer Turns Ratio			
Transformer Turns Ratio	N	15	1:1
Capacitor Value			
Capacitor Value	C	10	nF
Inductor Value			
Inductor Value	L	0.0001	H
Resistor Value			
Resistor Value	R	1	Ω

Template Legend

[Symbol] Input Voltage, User Input Variable
 [Symbol] Output Voltage, User Input Variable, do not alter
 [Symbol] Load Current, User Input Variable
 [Symbol] Resonant Frequency
 [Symbol] Switching Frequency
 [Symbol] Transformer turns ratio

LLC AC Circuit



Half Bridge LLC Resonant Converter Power Section Calculator

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